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Chilling the nerve, easing the pain?: A randomized clinical trial evaluating surgeon-administered bedside percutaneous cryoneurolysis for rib fracture pain

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BACKGROUND: A cornerstone of rib fracture management is multimodal pain control, which includes scheduled nonopioid analgesics, as-needed opioids, regional or neuraxial blockade, and surgical stabilization of rib fractures. However, adverse effect profiles and individual factors limit their use in all patients. Surgeon-administered, ultrasound-guided percutaneous cryoneurolysis performed at the patient bedside is a promising analgesic adjunct.

METHODS: We performed a prospective, randomized clinical trial assessing benefit and safety of surgeon-administered percutaneous cryoneurolysis to our existing multimodal rib fracture pain control bundle (standard of care [SoC]) for injured adults aged 18 to 64 years. Patients with fractured ribs 3 to 9 were randomized within 72 hours of admission to receive either surgeon-administered, ultrasound-guided percutaneous cryoneurolysis at the bedside and our multimodal pain control bundle, or SoC alone. Patients undergoing surgical stabilization of rib fractures were excluded. The primary outcome was pain score at discharge. Secondary outcomes included hospital length of stay, intervention-associated adverse events, morphine milligram equivalent (MME) use, Short Form (SF)-12, and McGill Pain Score (Patient Reported Outcomes), which were assessed at discharge and 1-, 3-, and 12-month intervals after discharge.

RESULTS: Forty-three patients were randomized: 24 in the intervention arm and 19 in the SoC arm, with 3 screen failures. The median age was 53 years (interquartile range [IQR], 45–60 years), 9 patients (23%) were female, the median Injury Severity Score was 13 (IQR, 10–17), and the median number of rib fractures was 6 (IQR, 4–8). Patients were well matched with no differences between groups. No intervention-associated adverse events were identified. Pain scores, MME use, and Patient Reported Outcomes were not different between groups at discharge, 1 month, 3 months, or 12 months.

CONCLUSION: Application of surgeon-administered, ultrasound-guided percutaneous cryoneurolysis at the bedside is safe but was not associated with reduced pain or MME use when compared with SoC. (*J Trauma Acute Care Surg.* 2025;00: 00–00. Copyright © 2025 The Author(s). Published by Wolters Kluwer Health, Inc. on behalf of the American Association for the Surgery of Trauma.)

LEVEL OF EVIDENCE: Therapeutic/Care Management; Level II.

KEY WORDS: Rib fracture; pain; neuralgia; disability; patient-reported outcome.

Rib fractures are common among bluntly injured patients and are thus frequently encountered by trauma surgeons, emergency physicians, and other clinicians who practice at trauma centers. Multimodal pain control is a cornerstone of rib fracture management, which commonly includes a combination of scheduled nonopioid analgesics, as-needed opioids, regional or neuraxial blockade, and surgical stabilization of rib fractures (SSRF), with the goal of reducing an injured patient's perception of pain related to fractured ribs, injured soft tissue, and irritated parietal pleura.[1,2] However, each of these modalities carries limitations and adverse effects that may impede their use in all patients.[3,4] Cryoneurolysis is another potential method of impeding nerve transmission from the injured chest wall to the central nervous system. Cryoneurolysis creates reversible anesthesia of the chest wall by freezing the intercostal nerves (−50°C to −70°C), which causes Wallerian degeneration of the nerve distal to the site of cryoneurolysis application.[5] The associated analgesic effect can last for 3 to 6 months as the nerve axon gradually regenerates along the perineural connective tissue (endo-, peri-, and epineurium), which remains intact after cryoneurolysis is applied.[5] Among modern studies evaluating cryoneurolysis for nontraumatic thoracic pathologies, the risk-to-benefit profile appears favorable. As such, cryoneurolysis may represent an important adjunct analgesic strategy to a multimodal pain control bundle for injured patients with rib fractures.[3]

Submitted: July 31, 2025, Revised: November 13, 2025, Accepted: November 17, 2025, Published online: December 29, 2025.

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This study was presented at the 84th annual meeting of the American Association for the Surgery of Trauma, September 10, 2025, in Boston, Massachusetts.

Supplemental digital content is available for this article. Direct URL citations appear in the printed text, and links to the digital files are provided in the HTML text of this article on the journal's Web site (www.jtrauma.com).

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DOI: 10.1097/TA.0000000000004885

In particular, handheld, ultrasound-guided, percutaneous cryoneurolysis is a particularly promising analgesic adjunct[4,6–9] and one that can be safely performed at bedside by trauma surgeons.[6,7] Its use may expedite trauma clinicians' ability to achieve early adequate pain control and allow for further individualization of multimodal pain control strategies.

Our hypothesis was that surgeon-administered, bedside, ultrasound-guided percutaneous cryoneurolysis is associated with decrease in numeric pain scores and morphine milligram equivalent (MME) use at discharge for injured patients with rib fractures. We thus performed a pilot, single-institution, prospective randomized clinical trial to assess the benefit of surgeon-administered, ultrasound-guided, percutaneous cryoneurolysis to our preexisting multimodal rib fracture pain control bundle.

PATIENTS AND METHODS

We performed a pilot, prospective, randomized clinical trial at our American College of Surgeons Level 1 trauma center to assess the addition of surgeon-administered, ultrasound-guided, percutaneous cryoneurolysis to our existing multimodal rib fracture pain control bundle (standard of care [SoC]) for injured adults aged 18 to 64 years. Bluntly injured patients with fractured ribs 3 to 9 were randomized within 72 hours of admission to receive either surgeon-administered, ultrasound-guided, percutaneous cryoneurolysis at the bedside and SoC, or SoC alone. Patients undergoing SSRF were excluded. Patients 65 years or older were excluded to because pf a competing, in-parallel clinical trial evaluating geriatric patients (NCT04482582). Additional exclusion criteria included an existing history of chronic chest wall pain, pregnancy, incarceration, inability to communicate or participate with investigators, inability to consent for procedure, pain score <5 with deep inspiration, radiographic evidence of metastasis to ribs, Glasgow Coma Scale score of <13, rib fractures located <3 cm from the transverse process, coagulopathy (international normalized ratio >1.5, platelet count <100 K), or inability to tolerate positioning for the procedure. Patients with any contraindication specific to percutaneous cryoneurolysis, including cold urticaria, cryofibrinogenemia, cryoglobulinemia, paroxysmal cold hemoglobinuria, and Raynaud's disease, were also excluded. Severity of rib fractures was allocated according to existing nomenclature using the radiology report; if a given rib was fractured in more than one location, the more severe fracture was recorded.[10] The primary outcome was pain score with deep inspiration at discharge. Deep inspiration was defined as an incentive spirometry volume of at least 500 mL. Secondary outcomes included hospital length of stay, intervention-associated adverse events, MME use, Short Form-12, and McGill Pain Scores (Patient Reported Outcomes), which were assessed at discharge and 1-, 3-, and 12-month intervals after discharge.

Patients were randomly allocated by the research coordinators to one of two treatment groups within 72 hours of admission: (1) SoC alone (Supplement Digital Content, Supplementary Data 1, http://links.lww.com/TA/F105) or (2) SoC with surgeon-administered, ultrasound-guided, bedside, percutaneous cryoneurolysis. All surgeons performing cryoneurolysis underwent the day-long, cadaver-based course provided and recommended by the device manufacturer. Once patients received cryoneurolysis, they were no longer eligible for any further neuraxial blockade. Computer-generated randomization was used by the clinical trials team who was uninvolved with patient care. Randomization occurred in a 1:1 ratio after study enrollment. No blinding was performed.

Patients randomized to percutaneous cryoneurolysis were positioned either in a lateral decubitus position or prone, received oxygen via nasal cannulate or facemask, and had standard American Society of Anesthesiologists (Shaumburg, Illinois) monitors applied (oxygen saturation, telemetry, blood pressure) in accordance with institutional policy. Patients were provided periprocedural fentanyl, as needed. The treatment sites were prepared with chlorhexidine gluconate. A 38-mm linear ultrasound transducer at 10 to 12 MHz was used to identify subject anatomy, and target ribs and nerves were identified in cross-sectional view. The costal angles 6 to 8 cm lateral to the spinous process (>3 cm from the transverse process) were marked. Local anesthesia was provided with 0.25% bupivacaine without epinephrine with weight-based limits prior to insertion of the cryoneurolysis probe. Treatment of nerves proceeded caudal to cranial. Cryoneurolysis was applied using the handheld 20-gauge probe, Iovera° Smart Tip 2190 (Pacira Biosciences, Inc., San Francisco, CA) in accordance with manufacturer guidelines and cryoneurolysis duration (106 seconds and −88°C). All accessible nerves of ribs 3 to 9 on the injured hemithorax were ablated. Postprocedural ultrasound and chest radiographs were obtained to identify any periprocedural development or interval worsening of new or existing pneumothoraces or hemothoraces. Adverse events were defined prior to study initiation as iatrogenic bleeding from injury to an intercostal artery, unintentional injury to an intercostal nerve, new pneumothorax or hemothorax, development of a neuroma, or a procedural site infection and were adjudicated by the study principal investigator in accordance with study protocol. Additionally, we evaluated for rib-specific readmissions. A rib-specific readmission was defined as a readmission associated with delayed hemothorax or pneumothorax, need for chest wall-specific pain control, pneumonia, or empyema.

Data for enrolled patients and screen failures were collected in REDCap. Reasons for screen failures were recorded and reported. A 50% enrollment interim analysis was performed and published in accordance with study protocol.[6] No patient or public involvement occurred during trial design. The sample size of 40 participants was determined based on our ability to fulfill established criteria for pilot studies and fall within the budget established by the sponsor.[11] The goal of this study was to assess feasibility, establish modern confidence intervals to inform a potential future multicenter Randomized controlled tria, and assess trend toward efficacy. Accordingly, outcome differences should be interpreted as exploratory and hypothesis generating. Data from this study were intended to provide adequate statistical support to determine if a signal toward benefit or harm existed, which would facilitate future multicenter trials.

Differences in pain scores between treatment groups were assessed using independent-samples *t* tests, with the exception of McGill Affective Pain Scores, which were compared using Wilcoxon rank-sum tests because of the presence of all zeros in the SoC group. Comparisons of numeric variables were performed using Wilcoxon rank-sum tests, and categorical variables were analyzed using χ^2 or Fisher's exact tests, as

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appropriate. Longitudinal changes in pain and daily MME were evaluated using Wilcoxon signed-rank tests. A linear mixed-effects model with random intercept and random slope was used to evaluate differential change in pain over time between groups. A two-sided p value of <0.05 was considered statistically significant, and corrections for multiple comparisons were not applied given the exploratory nature of this pilot randomized controlled trial. Analyses were conducted using SAS, version 9.4 (SAS Institute Inc., North Carolina).

This clinical trial was prospectively registered with clinicaltrials.gov (NCT05330611) on April 15, 2022. Consolidated Standards of Reporting Trials reporting guidelines were adhered to and the Consolidated Standards of Reporting Trials checklist is provided (Supplement Digital Content, Supplementary Data 2, http://links.lww.com/TA/F106). Written informed consent was obtained from all patients. No compensation was provided to patients for this study. Chest Wall Injury Society recommendations for reporting studies in chest wall injury were followed as applicable.[12] Institutional review board approval was obtained for this clinical trial.

RESULTS

Forty-three patients consented to participate in the study between February 2023 and January 2025, of whom 40 were enrolled (21 intervention, 19 SoC, and 3 screen failures) (Fig. 1). One intervention patient underwent the procedure outside of the 72-hour window because of an additional emergency surgery and scheduling difficulties. This patient is included in the primary analysis and excluded in a per-protocol sensitivity analysis. Two of three screen failures occurred because of patients' body mass index being prohibitive to anatomic visualization and one patient-reported pain resolution postrandomization. Patients were well matched, and there were no significant differences in demographics, injuries, rib-specific injuries, or other baseline measures between the two groups (Table 1). No patients had either concomitant sternal fracture or blunt cerebrovascular injury. No patients had clinical flail chest, as this population is offered SSRF in accordance with our institutional guidelines (Supplemental Digital Content, Supplementary Data 3, http://links.lww.com/TA/F107). All patients were receiving opioid pain medication preenrollment, and no patients received a thoracic epidural, paraspinal, or paravertebral block or continuous ketamine infusion preenrollment. After randomization and in discussion with the social work team, two intervention patients endorsed a history of active opioid use prior to injury and one SoC patient endorsed a history of prior opioid use (but current abstinence); none of these patients were excluded in accordance with study protocol.

No inpatient variables differed between the two groups (Table 2). Time from emergency department arrival to trial enrollment was equivalent between groups (intervention median [interquartile range, IQR], 1 (1–2) vs. SoC, 1 (1–1) days; $p = 0.191$). Among intervention patients, cryoneurolysis was performed within a mean (SD) of 4.6 (9.6) hours from enrollment. No intervention patients received neuraxial blockade after cryoneurolysis in accordance with study protocol, although two SoC patients (11%) received neuraxial blockade postrandomization. Two patients in the intervention group had a postprocedure chest tube place for a worsening of a preprocedurally existing pneumothorax or hemothorax. No procedure-related adverse events were identified. No patients in either arm required intubation or developed

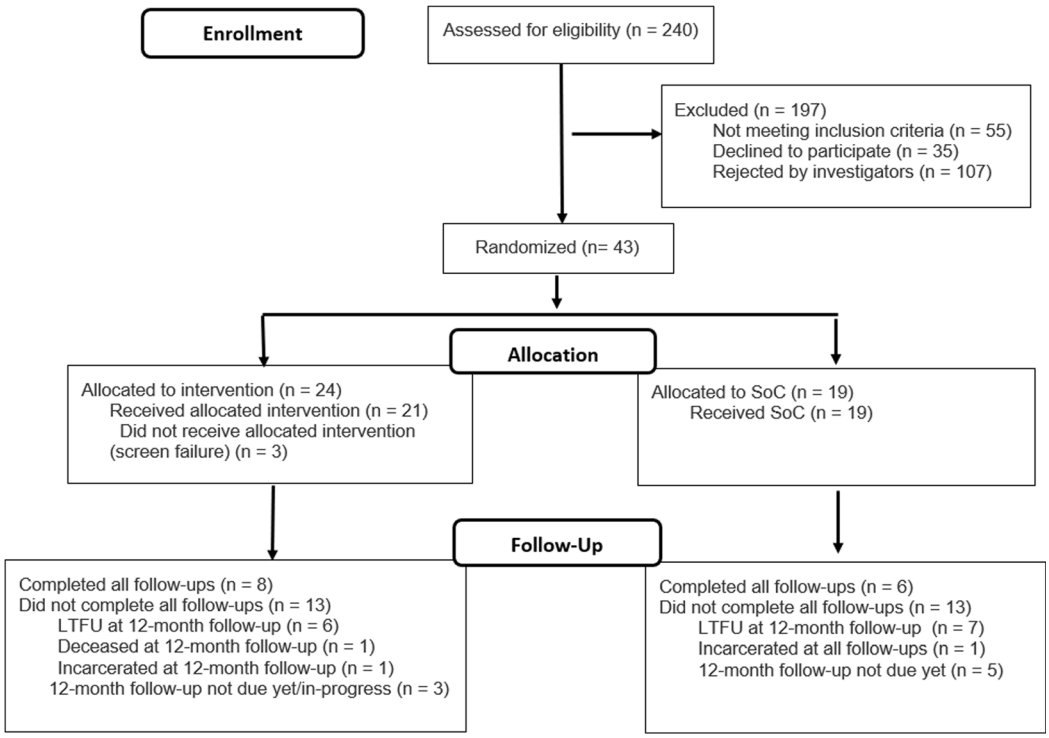

Figure 1. Consolidated Standards of Reporting Trials flow diagram for patient inclusion into study.

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TABLE 1. Demographic, Injury, Chest-Specific Injury, Comorbidity, Preenrollment Pain Therapy, and Preenrollment Pain Measures by Group

Characteristic	Intervention (n = 21)	SoC (n = 19)
Demographic variables		
Age, median (IQR), y	53 (49–58)	53 (37–60)
Female sex, n (%)	5 (24%)	4 (21%)
Unhoused, n (%)	2 (10%)	1 (5%)
Injury variables		
Mechanism of injury		
Fall	2 (10%)	3 (16%)
Motor vehicle	2 (10%)	3 (16%)
Motorcycle	3 (14%)	2 (11%)
Bicycle	9 (43%)	7 (37%)
Pedestrian vs. auto	4 (19%)	4 (21%)
Blunt assault	1 (5%)	0 (0%)
ISS, median (IQR)	13 (10–17)	13 (10–17)
Admission GCS, median (IQR)	15 (14–15)	15 (15–15)
Intracranial hemorrhage present, n (%)	1 (5%)	2 (11%)
Facial fracture present, n (%)	1 (5%)	1 (5%)
Spine fracture present, n (%)	4 (19%)	1 (5%)
Pelvic fracture present, n (%)	1 (5%)	2 (11%)
Long-bone fracture present, n (%)	3 (14%)	3 (16%)
Solid organ injury present, n (%)	3 (14%)	5 (26%)
More than one additional nonchest injury present, n (%)	2 (10%)	4 (21%)
Chest wall–specific variables		
Total no. rib fractures, median (IQR)	6 (4–8)	6 (3–8)
Laterality of rib fractures, n (%)		
Left only	9 (43%)	12 (63%)
Right only	11 (52%)	5 (26%)
Bilateral	1 (5%)	2 (11%)
Position of fractures, n (%)*		
Anterior	8 (38%)	8 (42%)
Lateral	18 (86%)	14 (74%)
Posterior	11 (52%)	7 (37%)
Severity of rib fractures, n (%)*		
Undisplaced	16 (76%)	13 (68%)
Offset	2 (10%)	0 (0%)
Displaced	14 (67%)	12 (63%)
Radiographic flail segment present, n (%)	3 (14%)	2 (11%)
Pneumothorax present, n (%)	13 (62%)	9 (47%)
Hemothorax present, n (%)	6 (29%)	1 (5%)
Clavicle fracture present, n (%)	6 (29%)	5 (26%)
Scapular fracture present, n (%)	2 (10%)	0 (0%)
Comorbidity variables		
COPD or asthma, n (%)	2 (10%)	0 (0%)
BMI >30 kg/m^2, n (%)	6 (29%)	4 (22%)
Current smoker, n (%)	5 (24%)	2 (11%)
Diabetes, n (%)	2 (10%)	3 (16%)
Prior rib fractures, n (%)	4 (19%)	2 (11%)
Preenrollment pain control variables		
Daily opioid equivalents in first 24 h of admission, median (IQR)	96.5 (42.0–135.4)	42.6 (26.0–147.5)
Continuous intercostal nerve block	1 (5%)	0 (0%)
Lidocaine GTT	4 (19%)	1 (5%)
Lidocaine patch	16 (76%)	17 (89%)
Gabapentin	18 (86%)	16 (84%)
NSAID	4 (19%)	5 (26%)
Acetaminophen	20 (95%)	19 (100%)
Preenrollment pain variables		
Highest pain score in first 24 h, median (IQR)	9 (8–10)	8 (7–10)
Median pain score in first 24 h	6 (5–7)	6 (4–7)
Cough, n (%)		
None	0 (0%)	1 (7%)
Weak	9 (64%)	3 (20%)
Moderate	3 (21%)	7 (47%)
Strong	2 (14%)	4 (27%)
Missing	7	4
Incentive spirometry, median (IQR), mL	1,500 (1,250–2,250)	1,625 (1,500–2,000)

*Based on patient having at least one fracture with this location or morphology.

BMI, body mass index; COPD, chronic obstructive pulmonary disease; GCS, Glasgow Coma Scale; GTT, continuous infusion; ISS, Injury Severity Score; NSAID, nonsteroidal anti-inflammatory drug.

either a pneumonia or a venous thromboembolic complication. Highest pain score upon discharge was similar between treatment groups. A per-protocol sensitivity analysis excluding the out-of-window intervention patient yielded similar results. Oral opioid usage was similar between groups upon discharge (intervention median [IQR], 60 [15.7–94.5] vs. SoC, 15 [0–60.0]; $p = 0.068$).

TABLE 2. Hospitalization and Discharge Destination by Group

Characteristic	Intervention (n = 21)	SoC (n = 19)	*p*
Hospitalization variables			
Hospital LOS, median (IQR), d	4 (3–7)	3 (2–4)	0.066
Admission to ICU, n (%)	3 (14%)	0 (0%)	0.233
Discharge variables			
Gabapentin prescribed, n (%)	19 (90%)	15 (79%)	0.398
Lidocaine patch prescribed, n (%)	17 (81%)	15 (79%)	1.000
NSAID prescribed, n (%)	8 (38%)	6 (32%)	0.666
Acetaminophen prescribed, n (%)	21 (100%)	16 (84%)	0.098
Highest pain score in 24 h before discharge, mean (SD)	7.48 (2.11)	6.47 (2.97)	0.223
Median pain score in 24 h before discharge, median (IQR)	5 (4–7)	5 (2–6)	0.334
Oral opioid usage day of discharge >0 MME, n (%)	19 (90%)	12 (63%)	0.060
Oral MME on day of discharge, median (IQR)	60.0 (15.7–94.5)	15 (0–60)	0.068

ICU, intensive care unit; LOS, length of stay; NSAID, nonsteroidal anti-inflammatory drug.

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Follow-up variables were similar between groups (Table 3). Among the 40 participants, 31 (16 intervention, 15 SoC) completed 1-month follow-up, 26 (14 intervention, 12 SoC) completed 3-month follow-up, and 14 (8 intervention, 6 SoC) completed 1-year follow-up. While there was no difference in pain at 1-month follow-up between intervention and SoC groups, McGill Present Pain Intensity (PPI) dropped significantly in the intervention group from discharge to 1-month follow-up (discharge PPI median [IQR], 7.00 (6.00–7.00) vs. 1-month follow-up PPI, 2.50 [1.00–5.50]; $p < 0.0001$), although this did not differ from the parallel drop in the SoC arm (Fig. 2). Distributions of categorical change in daily opioid usage did show a trend toward a higher proportion of intervention patients experiencing a decrease in daily opioid usage from discharge to 1-month follow-up (86% intervention vs. 61% SoC, $p = 0.055$). This can be partially explained by the higher proportion of SoC patients already off opioids at discharge (37% SoC vs. 10% intervention, $p = 0.07$). By 3 months, all patients were at home; all patients still using opioids noted decreased usage, and no patients were using lidocaine patches. The significant decrease in McGill PPI seen at 1 month among intervention patients did not persist at 3 months, among those with this information available (1-month follow-up PPI median [IQR], 3.00 [1.00–3.00] vs. 3-month follow-up PPI, 0.50 [0.00–4.00]; $p = 0.1914$). By 12 months, all patients had stopped using opioids (Fig. 3). There was no differential change in pain score over time between the

TABLE 3. Pain Medication Use and Patient-Reported Outcomes at 1-, 3- and 12-Month Follow-up

Characteristic	Intervention (n = 21)	SoC (n = 19)	*p*
1-mo Follow-up	**Intervention (n = 16)**	**SoC (n = 15)**	
Gabapentin still used, n (%)*	10 (48%)	5 (38%)	0.204
Lidocaine patch still used, n (%)*	3 (14%)	1 (6%)	0.609
NSAIDs still used, n (%)*	6 (29%)	2 (11%)	0.247
Acetaminophen still used, n (%)*	15 (71%)	9 (50%)	0.170
McGill Pain Score: present pain intensity, mean (SD)	3.19 (2.69)	2.07 (2.25)	0.220
McGill Pain Score: sensory, mean (SD)	2.19 (1.60)	1.53 (1.68)	0.277
McGill Pain Score: affective, mean (SD)	0.19 (0.54)	0 (0)	0.178
SF-12 PCS, median (IQR)	34.3 (23.0–42.3)	32.0 (24.7–50.0)	0.953
SF-12 MCS, median (IQR)	53.6 (48.1–57.4)	51.7 (49.4–57.7)	0.921
Using oral opioid, n (%)*	6 (29%)	2 (11%)	0.247
Daily total MME, median (IQR)*	0 (0–10)	0 (0–0)	0.201
Decrease in daily opioid usage from discharge among those with MME >0 at discharge, n (%)**	18 (95%)	11 (100%)	1.000
3-mo Follow-up	**Intervention (n = 14)**	**SoC (n = 12)**	
Gabapentin used, n (%)*	2 (10%)	0 (0%)	0.179
Lidocaine patch used, n (%)*	0 (0%)	0 (0%)	N/A
NSAIDs used, n (%)*	2 (10%)	1 (6%)	0.643
Acetaminophen used, n (%)*	5 (24%)	2 (11%)	0.303
McGill Pain Score: present pain intensity, mean (SD)	2.07 (2.76)	0.92 (1.51)	0.209
McGill Pain Score: sensory, mean (SD)	1.64 (2.27)	0.75 (1.22)	0.218
McGill Pain Score: affective, mean (SD)	0.50 (1.87)	0 (0)	0.396
SF-12 PCS, median (IQR)	40.9 (28.4–54.2)	54.6 (39.2–57.1)	0.117
SF-12 MCS, median (IQR)	56.0 (47.1–59.6)	51.8 (50.4–54.3)	0.292
Using oral opioid, n (%)*	2 (10%)	1 (6%)	1.000
Daily total MME, median (IQR)*	0 (0–0)	0 (0–0)	0.715
Decrease in daily opioid usage from last visit among those with MME >0 at 1 mo, n (%)†	6 (100%)	2 (100%)	N/A
12-mo Follow-up	**Intervention (n = 8)**	**SoC (n = 6)**	
Gabapentin used, n (%)*	1 (7%)	0 (0%)	0.326
NSAIDs used, n (%)*	1 (7%)	0 (0%)	0.326
Acetaminophen used, n (%)*	2 (15%)	0 (0%)	0.141
McGill Pain Score: present pain intensity, mean (SD)	0.75 (1.04)	0.83 (1.60)	0.908
McGill Pain Score: sensory, mean (SD)	0.63 (1.41)	0.33 (0.82)	0.66
McGill Pain Score: affective, mean (SD)	0 (0)	0 (0)	1.00
SF-12 PCS, median (IQR)	53.2 (42.8–55.6)	50.5 (24.0–55.9)	0.747
SF-12 MCS, median (IQR)	57.2 (52.1–61.3)	54.7 (49.4–56.0)	0.333

*Data for MME and nonopioid pain control were taken from charts even if patients did not follow up. Sample sizes after excluding missing data are as follows: 1-month and 3-month follow-ups: n = 21 for intervention, n = 18 for SoC; 12-month follow-up: n = 14 for intervention, n = 13 for SoC.

**n = 19 for intervention, n = 11 for SoC (n = 1 missing).

†n = 6 for intervention, n = 2 for SoC.

MCS, mental component score; N/A, not applicable; NSAID, nonsteroidal anti-inflammatory drug; PCS, physical component score; SF, Short Form.

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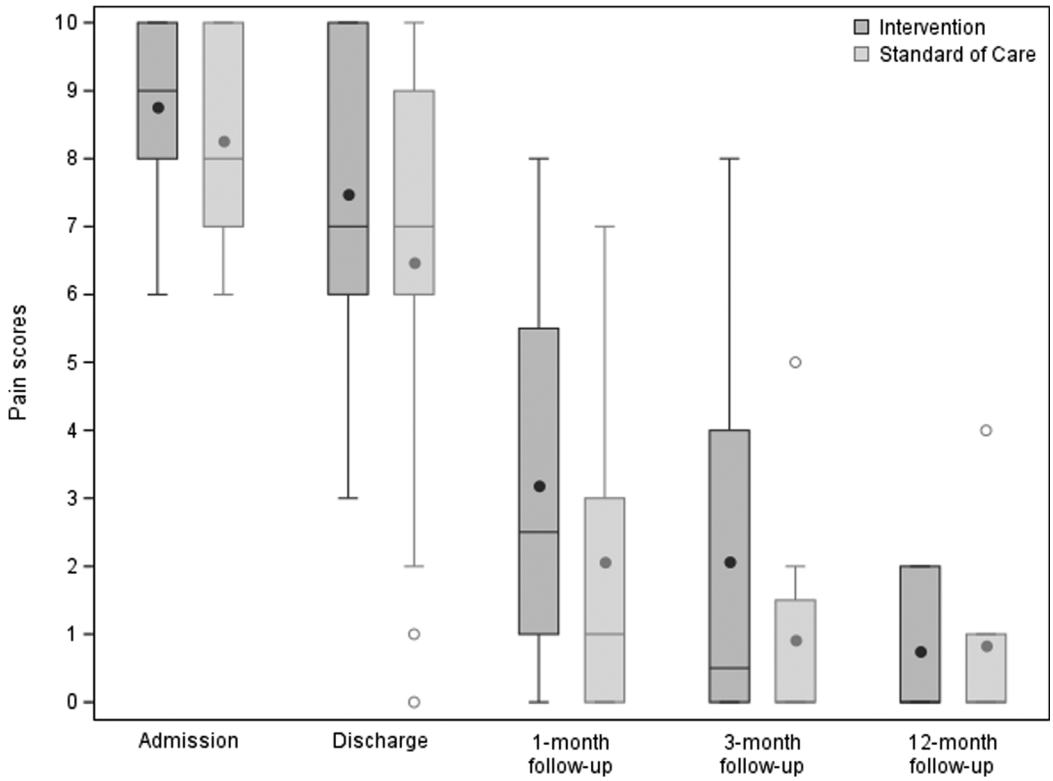

Figure 2. Distributions of pain from preprocedure to 12-month follow-up. The horizontal line within each box represents the median, while the lower and upper edge represent the 25th and 75th percentiles, respectively. The mean is shown by the dot inside the box. Whiskers extend to 1.5 times the IQR, and individual points beyond this range indicate potential outliers.

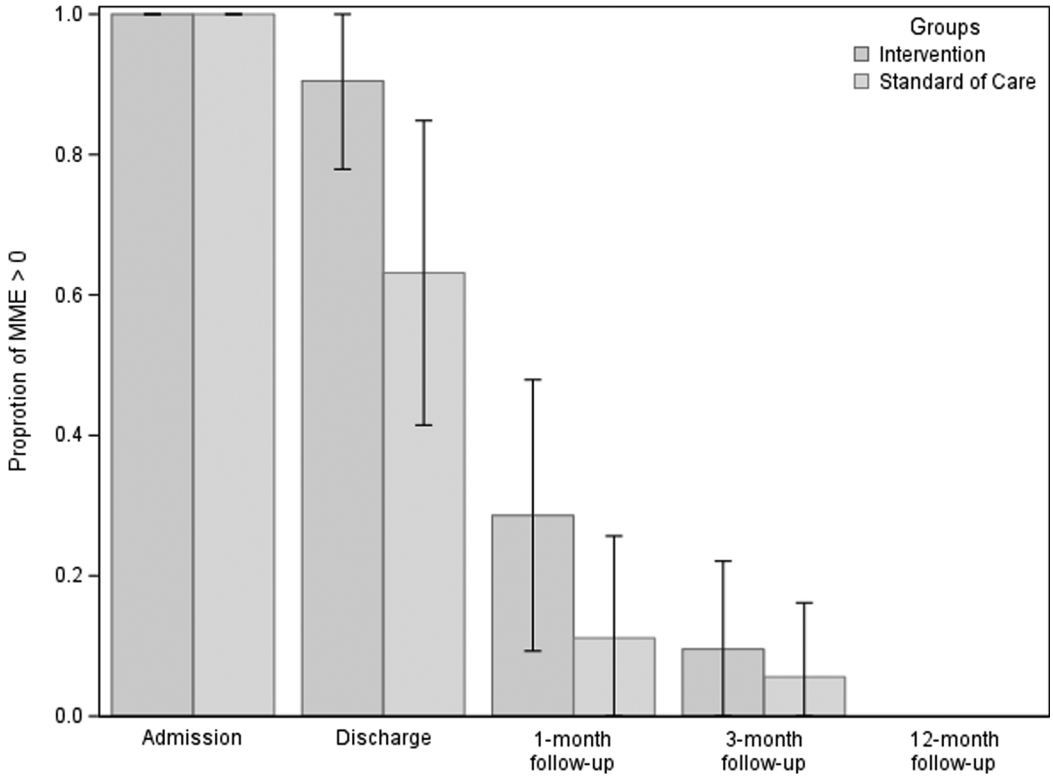

Figure 3. Proportion of patients on any opioids from preprocedure to 12-month follow-up. The error bar for each bar represents the 95% Wald confidence interval for the respective proportion.

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two groups. A per-protocol sensitivity analysis excluding the out-of-window intervention patient yielded similar results. No patient in either group experienced a rib-specific readmission over the 12-month follow-up period. No patient experienced winged scapula over the study period. There were no demographic, injury, or hospitalization characteristics that differed between those patients who remained in the trial compared with those who were lost to follow-up. Two participants were incarcerated at 1-year follow-up, and one died because of nontraumatic comorbidities before 1-year follow-up.

DISCUSSION

We demonstrated that surgeon-administered, ultrasound-guided, bedside, percutaneous cryoneurolysis can be performed safely, but it did not decrease pain scores or MME use when compared with the multimodal pain control bundle alone for patients with rib fractures. Our findings of safety are consistent with several other reported case series, trials, and systematic reviews, while the inability to demonstrate decreased pain or MME use was less consistent.[3,4,6–8] Among injured patients, safe implementation of percutaneous cryoneurolysis for pain control after rib fractures has been shown by a pain team and by interventional radiologists.[4,13] Our study adds to the growing body of literature that surgeons can also safely use this bedside technology.[6] In the operating room, use of intrathoracic cryoneurolysis at the time of SSRF has also been demonstrated to effectively reduce postoperative opiate consumption and favorably affect length of stay, although outcomes vary by institution.[14–19] Our study suggests that surgeons can safely apply the technology and conceptual framework used in the operating room at the bedside, yet it remains unclear if we should be adding percutaneous cryoneurolysis. This leads to the Jurassic Park paradox—just because we can be applying technology does not mean we should be applying this technology.

While the addition of percutaneous cryoneurolysis did appear to decrease pain in a similar fashion to SoC (i.e., numeric pain scores decreased in a similar fashion in both those who received the intervention and those who did not), it remains uncertain whether percutaneous cryoneurolysis should replace any therapeutic components in a multimodal pain control bundle, nor is it clear that percutaneous cryoneurolysis should be added at all. In fact, while not statistically significant, it is notable that more intervention patients were taking opioids on discharge and at greater volumes than the SoC patients. Thus, there is a potential that the intervention actually caused more pain than it prevented. Unfortunately, no universally agreed upon standards for minimum clinically important differences exist for rib fracture pain or MME use. Given how important these two variables are in evaluating interventions and therapies for this pathology, this is an unmet need that could facilitate interpretation of clinical benefit in further clinical trials.

Additionally, there are some limitations to application of this technology. Patients at the extremes of body habitus may be difficult to approach, as was the case with two of the screen failures in our study. It can be difficult to visualize relevant anatomy and appropriately administer the freeze in larger patients, and in thin patients, there is the risk of thermal injury at the skin because of proximity to the freeze zone. Furthermore, even with the limited power of our study results, there was not a marked discrepancy in overall pain scores, patient-reported outcomes, or MME use with the addition of percutaneous cryoneurolysis. Ultimately, it is unlikely that percutaneous cryoneurolysis alone will be adequate to fully address the totality of pain that many patients with rib fractures experience, although this does not eliminate its potential utility as part of a multimodal pain control bundle. As observed in patients with chronic pain, it may be difficult to understand the impact of any single intervention in a pain control bundle,[20] and any therapeutic intervention may be better viewed as an adjunct, rather than a supplement.

Future iterations on our study are encouraged to expedite time to intervention, as this may have an outsized impact on a patient's long-term recovery. No specialist is better suited to understand the impact of time on outcomes for the injured patient than the trauma surgeon. Across the spectrum of rib fracture management literature, time to pain control interventions has been shown to correlate with outcomes.[21–25] While our time to intervention achieved the standard set by study protocol, it is possible that earlier performance of percutaneous cryoneurolysis may have an even more substantive impact on a patient's pain symptoms. For example, percutaneous cryoneurolysis performed by the trauma team at bedside in the emergency department immediately upon identification of rib fractures could allow for further expedition of care of this injured patient population.

There are several limitations to this study. First, the study took place at a single American College of Surgeons Level 1 trauma center, and thus. its results may not translate to other settings. Second, no patients, surgeons, or other members of the treatment team were blinded to patients' randomization, potentially leading to overestimation of treatment effect through bias introduced by placebo effect in the intervention arm. It is possible that patients in the intervention arm experienced benefit solely because of perceived improvement in symptoms due to receiving a treatment that has no true therapeutic benefit, although the patient expects the treatment to work. Third, given the overall small sample size, patients lost to follow-up could have an outsized effect on reported outcomes, although there were no differences in demographic, injury, or hospitalization variables between those who received the intervention and those who did not. Fourth, this study was developed as a pilot effort to demonstrate safety and was underpowered to detect differences in our primary endpoint, potentially leading to type I error. Fifth, pain score with deep inspiration is an inherently subjective numerical measure as has been previously reported and could have been influenced by timing of pain medication among other variables that were not accounted for.[26] It is possible that using pain scores with movement could have either positively or negatively biased results, if applied. Sixth, as with any study or intervention reliant on ultrasound, it is possible for operator-dependent variability to exist, although this was minimal in our study given only three surgeons administered the intervention.[27,28] Seventh, as the principal investigator was the adjudicator of adverse events, it is possible that ascertainment bias was introduced. Finally, percutaneous cryoneurolysis was added to our multimodal SoC pain pathway rather than to its exclusion. As such, we are unable to comment upon treatment superiority, if present, of surgeon-administered, ultrasound-guided percutaneous cryoneurolysis. A subsequent multi-institution prospective clinical trial, ideally with sham

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control, would be ideal to help mitigate placebo effect and assess additive benefit, if any exists.

CONCLUSION

Application of surgeon-administered, ultrasound-guided percutaneous cryoneurolysis at the bedside is safe but was not associated with reduced pain or MME use when compared with SoC. Further research is needed before recommending percutaneous cryoneurolysis as a pain control adjunct for patients with rib fractures.

AUTHORSHIP

J.D.F. contributed in the study design, literature search, intervention performance, data collection, data analysis, data interpretation, writing, and critical revision. J.T.T. contributed in the data collection, intervention performance, writing, and critical revision. A.W.K. contributed in the intervention performance, data interpretation, and critical revision. S.W. contributed in the data interpretation and critical revision. S.G.-B. contributed in the data collection and critical revision. R.B.F. contributed in the data collection and critical revision. A.A.M. contributed in the data collection and critical revision. J.T.C. contributed in the data collection and critical revision. J.M.K. contributed in the data collection, data analysis, data interpretation, and critical revision. M.J.E. contributed in the data collection, data analysis, data interpretation, and critical revision. A.M.A. contributed in the data interpretation, project oversight, and critical revision.

ACKNOWLEDGMENTS

Pacira Biosciences was a partial funder for this work product. The remaining funds were provided by the Division of General Surgery at Stanford University School of Medicine.

DISCLOSURE

Conflicts of Interest: Author Disclosure forms have been supplied and are provided as Supplemental Digital Content (http://links.lww.com/TA/F104).

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